

EVOLUTION

Origin of life:

There are several theories that have been put forward trying to explain origin of life. Some of them are:

- Special creation

According to this theory; all living organisms were created by a supernatural being (GOD). It is based on Koran or Bible. It holds the following views that:

- i) The life was brought about by supreme being
- ii) The life forms were created in perfect form and have remained unchanged over time
- iii) It is based on faith, not disputable.

- Spontaneous generation

It holds that organisms were generated from non-living components e.g. maggots would spontaneously generated from fish meat left in an open container.

NB disapproved theory, not true.

- Steady state theory

It asserts that earth has no origin. It has always been able to support life. It has changed remarkably little if at all and that species on it had no origin.

- Cosmozoan theory

It holds that life may have originated elsewhere in the universe.

- Biochemical evolution

This is a scientific view of origin of life on earth. They based their explanation on what is termed as primitive earth and its atmosphere.

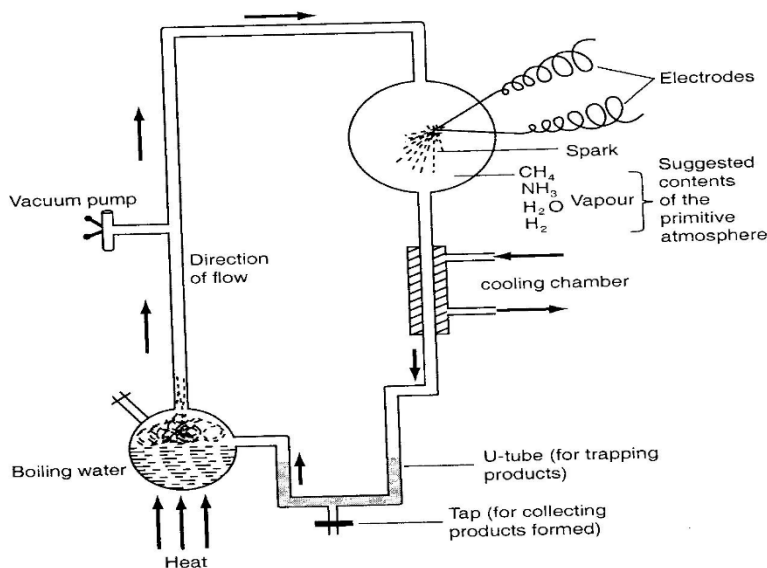
During that time conditions on that earth were adverse such as high temperature, lightning, ultra-violet rays.

These conditions acted as catalyst which favoured reaction of some elements resulting to simple molecules e.g. water H_2O , methane CH_4 , ammonia NH_3

Further combination of such compounds resulted to complex, self-replicating molecules resembling the present day DNA which forms genetic material of living cells.

Successive replication of these molecules and probably further development led to the formation of simple forms of living cells which resemble present day viruses and bacteria.

In 1953 S.L Miller was able to synthesis 4 of 20 known amino acids using apparatus that provided conditions such as high pressure and temperature, used a spark through water, methane, ammonia. Such conditions are believed to be present on universe during start of life.



It holds the following views:

- Life came into existence through combination of chemicals
- Initial life forms were simple and have changed to complex forms over time
- It has scientific evidences i.e. fossils

Chemical theory led to organic evolution as it hold a that initial life forms were simple and have changed in complexity over time.

➤ Organic evolution

Organic evolution is gradual change of organisms from simple forms to complex forms over a period of time. This theory has many evidences:

- a) Fossil records
- b) Geographical distribution
- c) Comparative anatomy
- d) Comparative embryology
- e) Cell biology
- f) Comparative serology

Fossil records

Study of fossils is called palaeontology. Fossils are remains of organisms that existed long time ago that have been preserved in plant resins and sedimentary rocks by chance.

These remains include bones, artifacts, and teeth. Most common method of fossil formation is petrification i.e. changing into rocks.

Fossils are important because:

- i) They show an evidence of the type of organisms that existed long time ago.
- ii) They show gradual change from one form to another over a long period of time.
- iii) They show evolutionary relationship between organisms.
- iv) They are used to construct evolutionary history of organisms such as humans.

The age of fossils is determined by using radioactive dating i.e. carbon dating.

Fossils have some weakness i.e.:

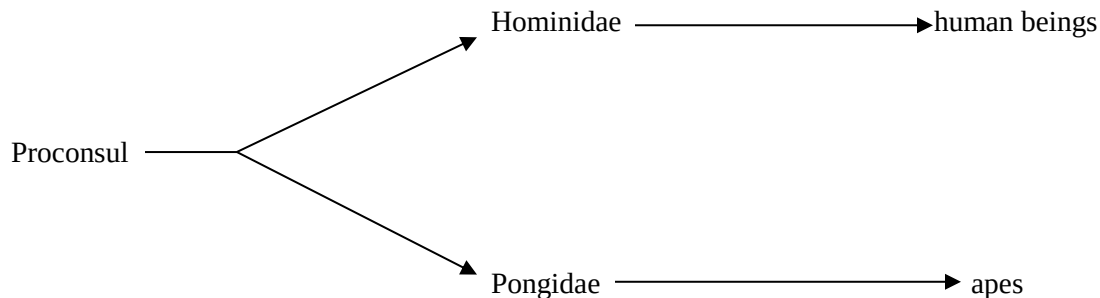
- There are missing links due to complete decomposition of organism or scavenging.
- Distortion of parts during sedimentation which give wrong impression of the structures
- Destruction of fossils by geological activities e.g. faulting, mass movement.

Fossil records of man

Darwin proposed that man evolved from ape like creatures. The earliest primate stock from which man evolved from was proconsul that lived 20 million years ago.

Proconsul evolved into two:

- Hominidae which man belong
- Pongidae which anthropoid apes like gorilla, chimpanzee belongs.



Best known hominides (4 million years ago) belonged to genus ***Australopithecus*** found in Olduvai gorge, Tanzania and Ethiopia.

Zinjanthropas boisei (1.8 million years ago) was found in Olduvai gorge.

Genus Homo arose 2.5 million years ago. Oldest known species is ***Homo rudolfensis*** dated 2.5 million years. ***Homo habilis*** (handy man) dated 2 million years made tools such as hand axes, scrapers. It became extinct 1.5 million years ago.

Homo erectus or “erect man” emerged from H. habilis in addition to tool making they had communal life and work.

Homo erectus were different from man by having prominent brow ridge over eyes, small stature and brain capacity between 775-1200 cm³

Homo neanderthalensis dated 300,000 years old was discovered in Germany.

Homo sapiens or rationalizing man is thought to have risen 130,000 years ago. The present varieties of man are seen to have risen about 10,000 years ago.

Homo sapiens are characterized by:

- High intellectual capacity
- The ability to vocalize sound i.e. communicate through language
- Express moral judgment
- Locomotion by bipedal
- Reduced incisors and canine teeth unsuitable for killing prey and self-defense.
- Rational / develop ideas
- Opposable thumb on fore limbs for manipulation
- Forward eyes for 3 dimensional view hence able to judge.

Main difference between ape and man

Ape	man
Quadrupedal gait	Bipedal gait
Small brain	Enlarged brain
Pelvis narrow and elongated	Pelvis broad and flattened
Hind limbs has opposable toe for grasping branches	Non-opposable toe on foot used for stability on ground
Large incisors and canine useful in defense and killing prey	Reduced incisor and canine teeth unsuitable for killing prey and self-defense

Geographical distribution

It is believed that long time ago, all continents form one large land mass (Pangaea). The single land mass broke into two, northern continent (Lauresia) and southern (Gondwanaland). These further broke into present day continents by process of continental drift.

Closely related organisms were separated and isolated from each other; leading to formation of subsequent different species (speciation).

Different species of organisms inhabit different continents. Although certain climatic conditions are similar in the same latitudes at different continents, their flora and fauna are not identical e.g.

- ❖ Monkeys with long prehensile tails live in Amazon forest in South America while in African forests there are short tailed monkeys.
- ❖ The llamas of the Amazon forest and modern camels of Africa and Asia are thought to have common ancestry in N. America but migrated and become isolated thus evolved into different species.
- ❖ Australia drifted before evolution of placental mammals (placental era); thus organisms found there have no placenta. Young ones are born prematurely and placed in a pouch (marsupials)

- ❖ Isolated oceanic islands evolved their own characteristic plants and animals e.g. Galapagos Island has both giant iguana lizard which are terrestrial and marine.
- ❖ Cat family e.g. cheetah in Africa, tiger of Asia and jaguar of South America developed from one species.

Comparative anatomy

This is comparing the form and structures of different organisms. Some have similarities in structure and are believed to share common origin.

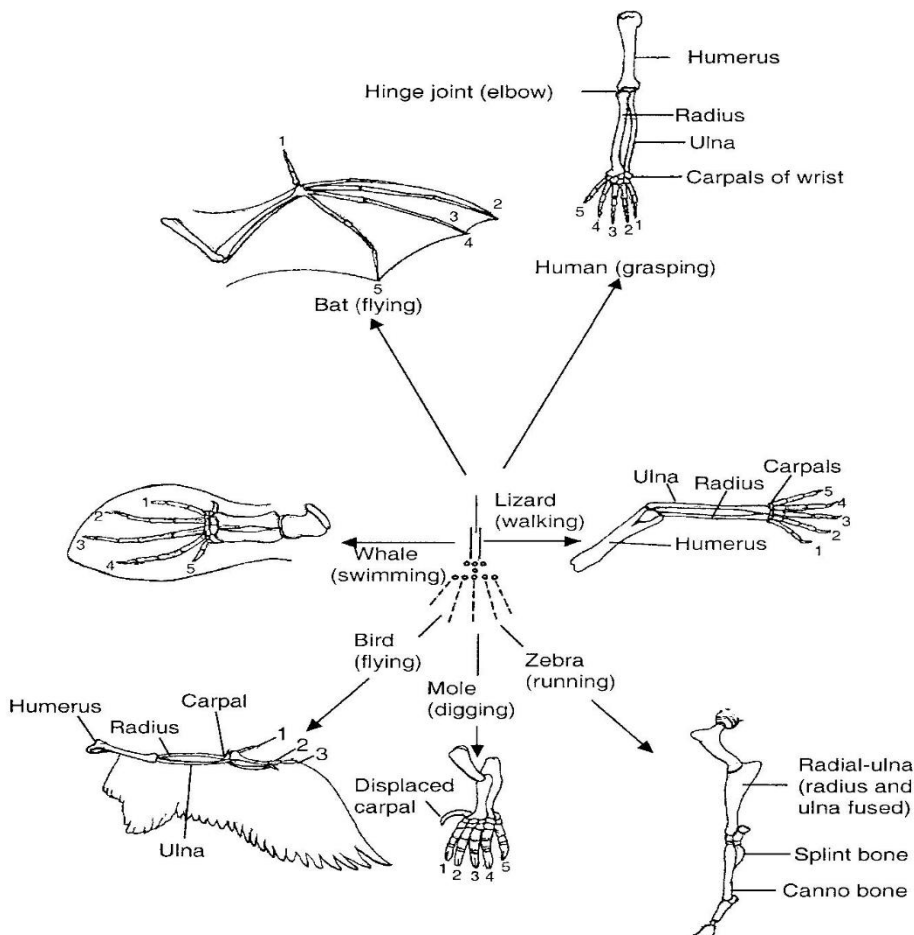
These structures can be grouped into:

- ✓ Homologous structures

These are structures with common embryonic origin but modified within the course of evolution to perform different functions in order for them to exploit different ecological niches leading to **divergent evolution**.

The evolutionary process in which forms originating from one common ancestry may be modified to adapt to different ecological niches is known as **adaptive radiation**. For example:

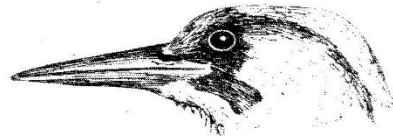
- Pentadactyl limbs in vertebrates



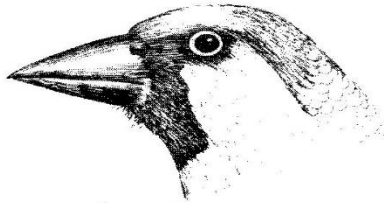
- Beaks of birds



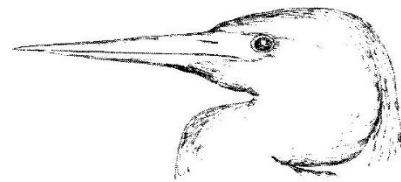
A flesh eater



A worm and insect eater

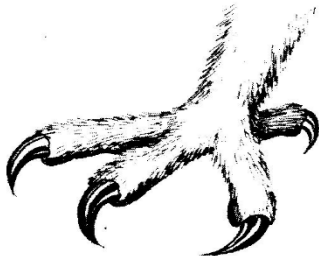


A grain eater

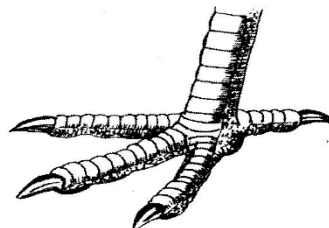


A fish eater

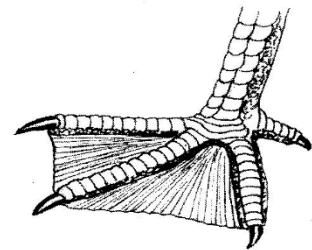
- Feet in birds



Foot of a flesh eater



Foot of a grain/insect eater



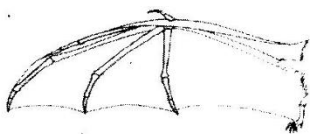
Foot of a fish eater

- ✓ Analogous structures

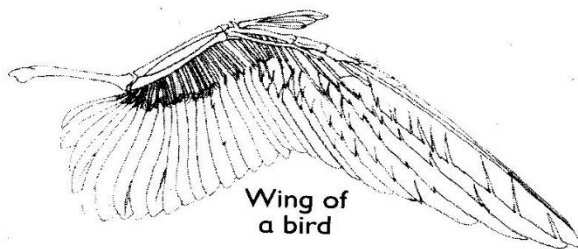
These are structures with different embryonic origin but modified in process of evolution to perform similar functions for adaptation and exploitation of the same ecological niche.

This lead to type of evolution called **convergent evolution** examples:

- Wings of birds, bats and insects



Wing of a bat



Wing of a bird



Wing of an insect

Birds originate from endoskeleton while insects from exoskeleton

- Eye in humans and that of octopus
- ✓ Vestigial organs

Structures reduced in size and have become functionless in the course of evolution but were functional in ancestral forms e.g.

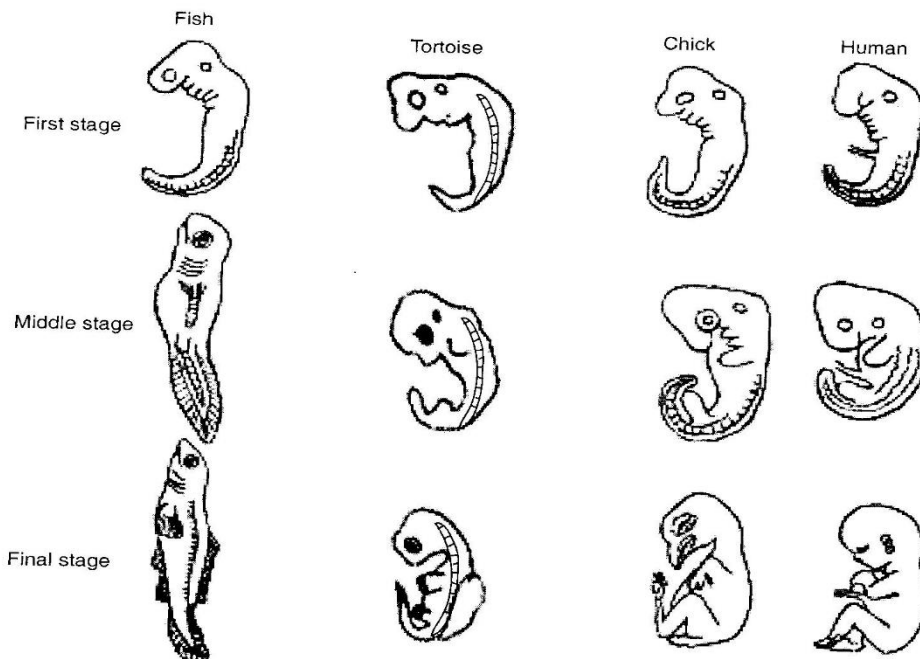
- Reduced wings in kiwi, ostrich
- Appendix, coccyx in humans
- Nictitating membrane in human eyes

Comparative embryology

This is the study of formation and development of an embryo. Embryos of different vertebrates groups have been found to have similar morphological features during their early stages of development suggesting common ancestry.

This theory called recapitulation.

The closer the resemblance between embryos in early stages the closer the phylogenetic relationship of the organisms.



Cell biology

Cells of all organisms show basic similarities in their structure and functions. All these contains cell membrane, ribosomes, Golgi bodies, mitochondria.

They also have some biological chemicals in common e.g. ATP & DNA

These strongly indicate that all cell types have common ancestral origin.

There are also some difference between plants and animals i.e. plants have cellulose cell wall, chloroplast, cell sap which are absent in animals. These differences indicate that plants and animals have common ancestry but evolved along different lines.

Comparative serology

Analysis of blood proteins and antigens show phylogenetic relationship. Species more phylogenetically related contain similar blood proteins.

The amount of immunological reactions (antigen- antibody reactions i.e. agglutination) can be used to show relationship between organisms.

The greater level of precipitate; the closer phylogenetic relationship among organisms.

Mechanism of evolution

This mechanism is based on two theories:

a) Lamarck's theory

The theory proposed that when environment demanded the use of a particular structure on an organism, the organism responded by developing it.

And if environment does no longer need a certain structure, the structure degenerate with time and disappears.

This result to **use** and **disuse** of structures; that led to changes on organism with time. It holds that such demandable characteristics were transmitted or passed to their offspring's and subsequent generation hence leading to emergence of new species.

Example:

Lamarck proposed that the present day giraffe with long neck resulted from ancestral short neck giraffes through inheritance of acquired characteristics of long neck.

Long neck trait resulted when food near the ground got diminished due to competition, so only food available were those on high up trees. Giraffes on straining to reach food the neck got elongated and became more competitive thus outlining the short necked.

This advantageous characteristic was then transmitted to next generation.

Weakness of this theory

Lamarck's explanation was right about dictation of environment on lifestyle of organisms but this changes were only phenotypic.

These changes does not affect the genotype of organism thus cannot be inherited.

b) Darwin's theory of natural selection

Darwin in his book (*The Origin of Species 1859*) put forward his explanation based on a concept called natural selection.

The main features are as follows:

- ❖ It is observed that variation occur within members of population through sexual reproduction; the **variations** are transmitted to successive offsprings. Some of such variations are advantageous while others are disadvantageous.
- ❖ Always the number of offsprings by far outnumbers the parent generation but few survive to adulthood and are able to reproduce. This is due to environmental pressures such as predators diseases and competition therefore members of a population remain relatively constant. This means members of the population are constantly competing with each other in an effort to survive. This is referred to as **struggle for existence**.
- ❖ In struggle for existence individuals with advantageous variations are well adapted to environment and are selected hence survive and reproduce. Those who are poorly adapted (disadvantageous variations) perish before transmitting genes to offsprings. This is called **survival for the fittest**.
- ❖ Natural selection is a chance occurrence which selects individuals or species that are best suited to an environment to continue surviving and reproduce; while eliminating those less suitable from existing.

The selected individuals transmit their advantageous variations to their offsprings.

The gradual accumulation of small variations from generation to generation over a period of time leads to emergence of new species.

The variation that Darwin regarded as raw materials for process of evolution have been traced to changes that occur at the level of the gene or chromosome; these changes are called **mutation**.

Organisms with beneficial mutations are able to pass them to their offsprings. Transmissions of such mutations through many generations lead to emergence of new species.

Natural selection in action

Natural selection in action results in formation of new species which takes many years. So it has been hard to see it taking place.

What is seen is varieties within any species. Varieties offer evidence of evolution in progress. They include:

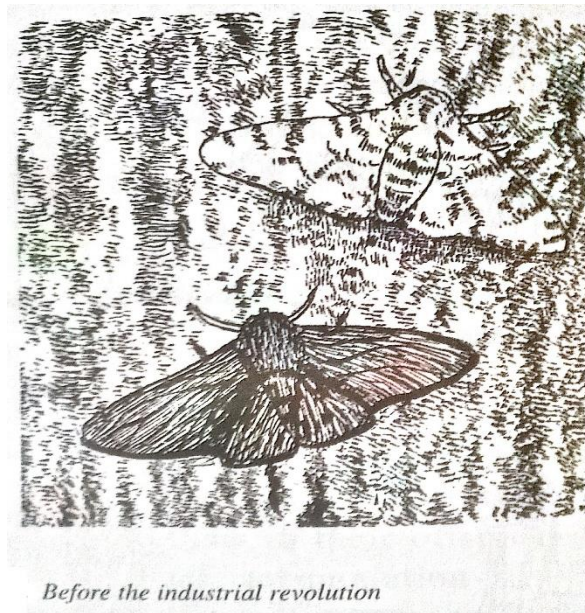
- a) Industrial mechanism
- b) Resistance to drugs, pesticides, antibiotics

Industrial mechanism

Consider peppered moth (*Biston betularia*)

It is found in many parts of England. It normally rests on tree trunks and branches. It exist in two varieties a normal variety or wild type that is speckled white and a mutant variety black (melanic form).

Before industrial revolution almost all peppered moths were speckled white variety.



They were well camouflaged against the tree trunks within similar colouration.

Since 1940's increasing numbers of melanic forms started to appear in areas surrounding industrial areas. This is because the trees were darkened by soot from industries; this helped the black moths camouflaged by dark background protecting them from predators.

White moths were usually picked and eaten by birds. So soot free areas, speckled white form predominate.

This case may well represent a possible transition towards speciation but the moment the varieties represent polymorphic forms within a species. They don't represent two different species.

Sickle cell trait

Sickle cell trait is another example of natural selection in action

There is high frequency of this mutant gene (sickle cell) where malaria is very prone.

This is because those who are heterozygous $HB^A HB^S$ have immunity resistance to malaria. They are known as heterozygous advantage and are selected for by nature.

Resistance to drugs

Drugs, antibiotics and pesticides are chemicals used to kill disease causing micro-organisms and pests respectively. Antibiotics and pesticides are rarely 100% effective. Some pathogens and pests usually survive and reproduce even when exposed to antibiotics or pesticides.

This resistance shows that within population some individuals passes the gene for resistance or acquire it through mutation hence survive chemicals.

